

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 13.11.2019

Day: Wednesday

Time: 10:00 am to 10:30 am

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : BPT- November 2019
Semester: 4 th
Section : I
Course Code : PT 257
Course Name: General Surgery

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes. Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs) (Encircle the correct answer)

(20 × 1 = 20)

1. First step in trauma is _____.
 - a. Blood transfusion
 - b. IV fluid
 - c. Reconstruction
 - d. Maintenance of airways
2. Graft from sister to brother is called _____.
 - a. Isograft
 - b. Allograft
 - c. Autograft
 - d. Heterograft
3. Most common complication of immunosuppressant is _____.
 - a. malignancy
 - b. graft rejection
 - c. infection
 - d. Thrombocytopenia

(P.T.O.)

4. Normal pH is _____.
 - a. 7.25-7.35
 - b. 7.35-7.45
 - c. 7.55-7.65
 - d. 7.45-7.55
5. Which of the following is best method to assess to fluid intake?
 - a. Urine output
 - b. CVP
 - c. Pulse
 - d. BP
6. Which of the following is best indicator for blood transfusion?
 - a. urine output
 - b. Hematocrit
 - c. color of skin
 - d. clinical examination
7. All of the following are the complications of massive blood transfusion except _____.
 - a. hypokalemia
 - b. hypothermia
 - c. hypomagnesaemia
 - d. hypocalcaemia
8. Shock is clinically best assessed by _____.
 - a. Urine output
 - b. CVP
 - c. BP
 - d. Hydration
9. Hemorrhage leads to _____.
 - a. septic shock
 - b. neurogenic shock
 - c. hypovolaemic shock
 - d. cardiogenic shock
10. Thromboangitis obliterans is also known as _____.
 - a. Giant cell arteritis
 - b. Buerger's disease
 - c. Kawasaki disease
 - d. Raynaud's disease
11. Contraindication to Lung transplantation is _____.
 - a. Pulmonary arterial pressure is >55 mmHg.
 - b. PaCO₂ is more than 55 mmHg
 - c. FEV1 is less than 25% of the predicted normal value
 - d. Malignancy in last 2 years.
12. Hematuria is _____.
 - a. Blood in sputum
 - b. Blood in Urine
 - c. Blood in expectoration
 - d. Blood in vomit
13. Pus collection in pleura is called _____.
 - a. Hydrothorax
 - b. Pleural effusion
 - c. Empyema
 - d. Emphysema

14. Which of the following is NOT used as a graft to bypass coronary arteries?
- Radial artery
 - Femoral artery
 - Saphenous vein
 - Internal mammary artery
15. Removal of the broncho-pulmonary segment of the lung is called_____.
- Lobectomy
 - Wedge resection
 - Segmental resection
 - Pneumonectomy
16. In which of the following hernia, the blood supply of its content is impaired?
- Reducible
 - Irreducible
 - Obstructed
 - Strangulated
17. _____ is not a type of 'distributive shock'.
- Septic
 - Endocrine
 - Neurogenic
 - Anaphylactic
18. In Raynaud's disease color changes occurs in which pattern?
- white, blue, red
 - red, blue, white
 - blue, red, white
 - white, red, blue
19. Which of the following is acyanotic congenital heart disease?
- Atrial septal defect
 - Tetralogy of fallot
 - Transposition of great arteries
 - a & b
20. Rodent ulcer has_____ edges.
- everted
 - undermined
 - rolled
 - sloping

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B.P.T. Examination

November 2019

PT257 General Surgery

Date: 13.11.2019

Time: 10:30 am to 01:00 pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. A 68 year old male Madhav Jagad who is a chronic smoker is complaining of paroxysmal coughing and protrusion of viscera through his umbilicus. His BMI is 39kg/m². According to you what is the cause/s for the protrusion of the viscera?
- b. In which conditions drainage tubes and system are used? Name any 4.
- c. Differentiate between arterial occlusion and venous occlusion. (any 4)
- d. By which routes cancer spreads in our body? Enumerate them.
- e. What is appendectomy and by which incision commonly it is performed?
- f. What is homeostasis? Enumerate its components.
- g. What are the indications for nephrectomy? (any 4)

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Which are the factors affecting wound healing? Describe any 3 of them.
- b. Can an 85 year old Karshandas Joshi who died due to Lung carcinoma be a heart donor? Justify your answer. Describe the criteria for determining brain death. (2.5+2.5)
- c. What are the complications of thoracic surgery? Describe any 4 of them.
- d. Do you agree with the statement that some congenital cardiac conditions require palliative procedure before curative surgery/ies are performed”. Justify your answer.
- e. A 6 year old boy who is burned due to a fire accident is brought to the burns unit. How to determine the percentage and depth of burns for this patient? Explain it.

Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. Explain types of anesthesia and their complications.(5+5)

(P.T.O.)

b. Describe the causes, pathophysiology and surgical management of Carcinoma Breast.

(2.5+2.5+5)

c. A 60 year old Shilpaben Kumbhar complains of pain in both the lower limbs while walking for 10 minutes which relive upon rest for 3 minutes after which she resumes her walking again. She is suffering from diabetes since past 5 years. (2.5+2.5+5)

- i. Name the clinical feature described by the patient.
- ii. Which condition patient is most likely to suffer from?
- iii. Which risk factors are associated with the development of the clinical feature described by the patient? Describe any 5 of them.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B.P.T. Examination

November 2019

PT257 General Surgery

Date: 13.11.2019

Time: 10:00 am to 01:00 pm

Maximum Marks: 50

Answer Key

Section I

Sr. No.	Answer
1.	d
2.	b
3.	c
4.	b
5.	a
6.	b
7.	a
8.	a
9.	c
10.	b
11.	d
12.	b
13.	c
14.	b
15.	c
16.	d
17.	b
18.	a
19.	a
20.	c

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. A 68 year old male Madhav Jagad who is a chronic smoker is complaining of paroxysmal coughing and protrusion of viscera through his umbilicus. His BMI is 39kg/m². According to you what is the cause/s for the protrusion of the viscera?
- The causes for the protrusion of the viscera are coughing and obesity.
- b. In which conditions drainage tubes and system are used? Name any 4
- In Pneumothorax, Pleural effusion, pericardial effusion, Empyema, Hydrothorax etc. drainage tubes and system are used.
- c. Differentiate between arterial occlusion and venous occlusion.

	Arterial Disease	Venous Disease
Skin	cool or cold, hairless, dry, shiny, pallor on elevation, rubor on dependent area	warm, though, thickened, mottled, pigmented
Pain	sharp, stabbing, worsens w/ activity and walking, lowering feet may relieve pain	aching, cramping, activity and walking sometimes help, elevating the feet relieves pain
Ulcers	severely painful, pale, gray base, found on heel, toes, dorsum of foot	moderately painful, pink base, found on medial aspect of the ankle
Pulse	often absent or diminished	usually present
Edema	infrequent	frequent, esp. at the end of the day and in areas of ulceration

d. By which routes cancer spreads in our body? Enumerate them.

- 1 Transcoelomic
- 2 Lymphatic spread
- 3 Hematogenous spread
- 4 Canalicular spread

e. What is appendectomy and by which incision commonly it is performed?

An appendectomy is the surgical removal of the appendix using following incisions:

1. McBurney's incision, also known as grid iron incision
2. Lanz incision
3. Rutherford morrison
4. Paramedian incision

f. What is homeostasis? Enumerate its components.

- Homeostasis is the state of steady internal physical and chemical conditions maintained by living systems. All homeostatic control mechanisms have at least three interdependent components for the variable being regulated: a receptor, a control centre, and an effector.

g. What are the indications for nephrectomy?(any 4)

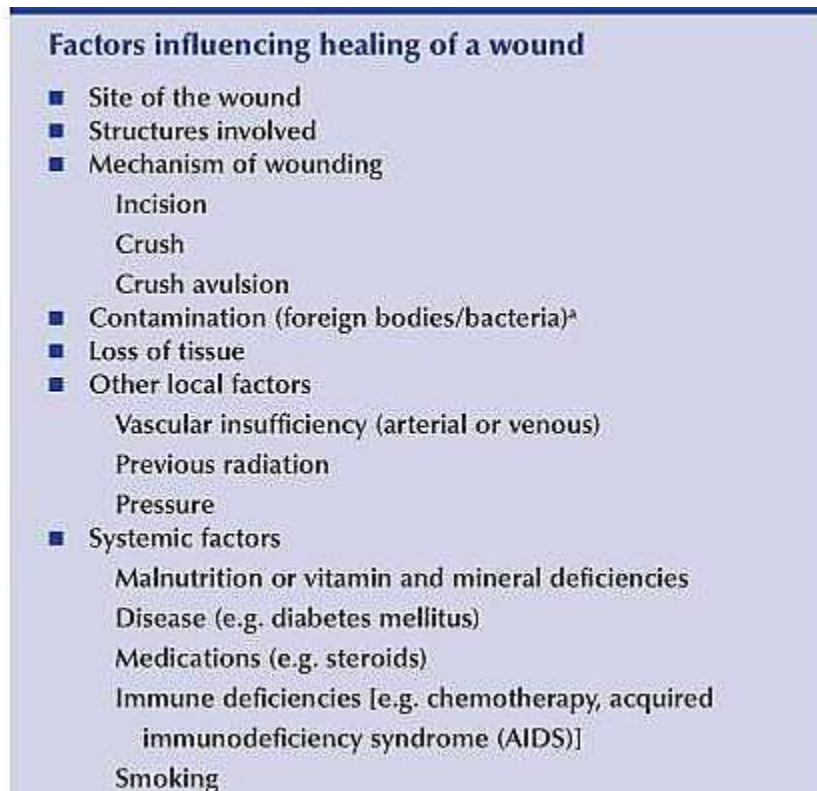
Indications of nephrectomy

- ▶ Benign conditions :
 - irreversable damaged kidney due to:
 - chronic infection .
 - obstruction .
 - trauma .
 - renal TB .
 - renovascular hypertension .
 - polycystic kidney disease .
 - transplant .
- ▶ Malignant : renal cell carcinoma , chromophobe carcinoma

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words)

(4 × 5 = 20)

- a. Which are the factors affecting wound healing? Describe any 3 of them.



- b. Can an 85 year old Karshandas Joshi who died due to Lung carcinoma be a heart donor?

Justify your answer. Describe the criteria for determining brain death. (2.5+2.5)

No he cannot be a heart donor because his age is 85 years and he died due to lung carcinoma.

Criteria for determining brain death are:-

1. Mechanism of brain injury is sufficient to account for irreversible loss of brain function
2. Absence of reversible causes of CNS depression
3. CNS depressant drugs
4. Hypothermia (<32°C [85°F])
5. Hypotension (MAP <55 mmHg)
6. Absence of neuromuscular blocking drugs that may confound the results of the neurologic exam
7. No spontaneous movements, motor responses, or posturing
8. No gag or cough reflexes
9. No corneal or pupillary light reflexes
10. No oculovestibular reflex (cold caloric)

Confirmatory Tests

- Apnea test for minimum of five minutes showing:
- No respiratory movements
- PCO₂ >55 mmHg
- pH <7.40
- No intracranial blood flow

c. What are the complications of thoracic surgery? Describe any 4 of them.

Respiratory

- infection of the lung
- consolidation / collapsed
- pneumothorax
- broncho-pleural fistula

Circulatory

- DVT
- Cardiac arrhythmia
- Hemorrhage

Wound:-

- Infection
- Failure to heal
- Adherent scar

Joint stiffness:-

- Shoulder joint
- Thoracic spine
- Costo-vertebral joints

Muscle weakness

Postural deformity

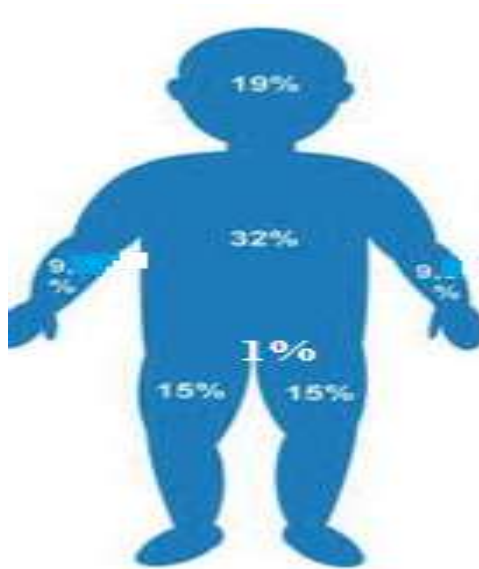
d. Do you agree with the statement that some congenital cardiac diseases require palliative procedure before curative surgery/ies are performed". Justify your answer.

Yes, I agree because the palliative procedure:-

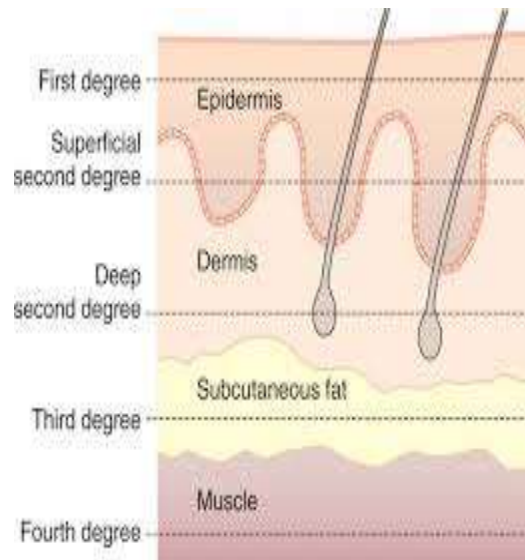
- Provides symptomatic relief,

- ii. Makes cardiac malformation more tolerable,
 - iii. Providing an improvement in the patient's condition and
 - iv. Permitting growth until the child has complete correction.
- e. A 3 year old boy who is burned due to a fire accident is brought to the burns unit. How to determine the percentage and depth of burns for this patient? Explain it.

Figure of nine can be used to determine the percentage of burns for this patient:



Burns are classified on the basis of depth of burns which can be utilized for this patient:

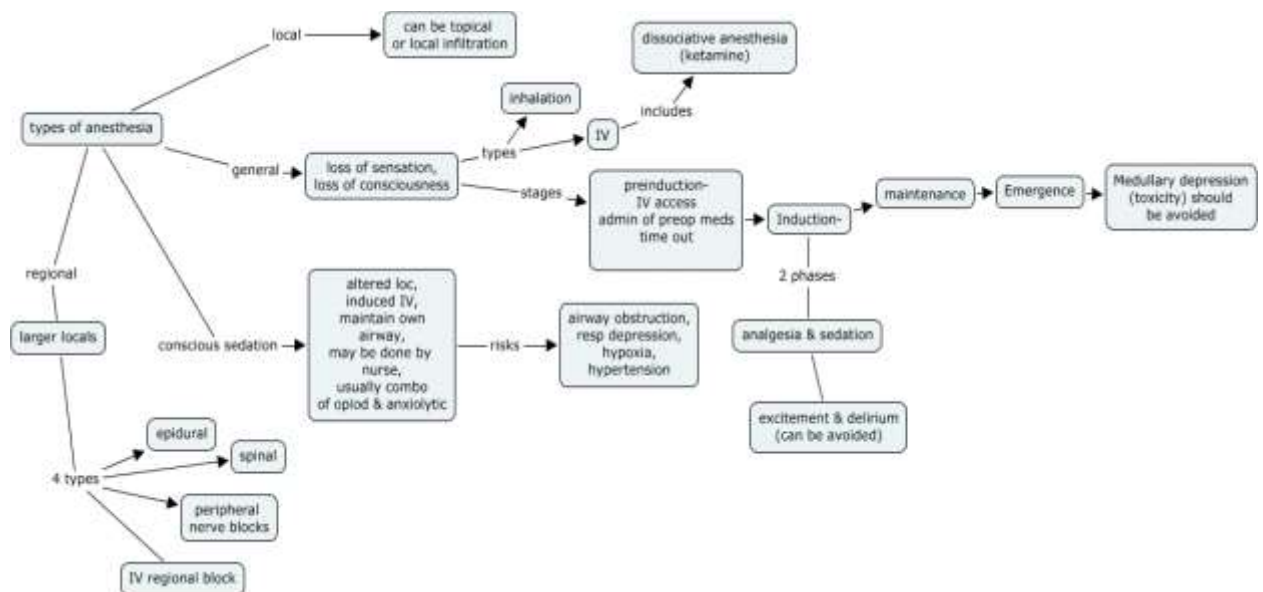


Q. 3. Essay (2 out of 3)

(2 × 10 = 20)

- a. Explain types of anesthesia and their complications.

Types of Anesthesia:

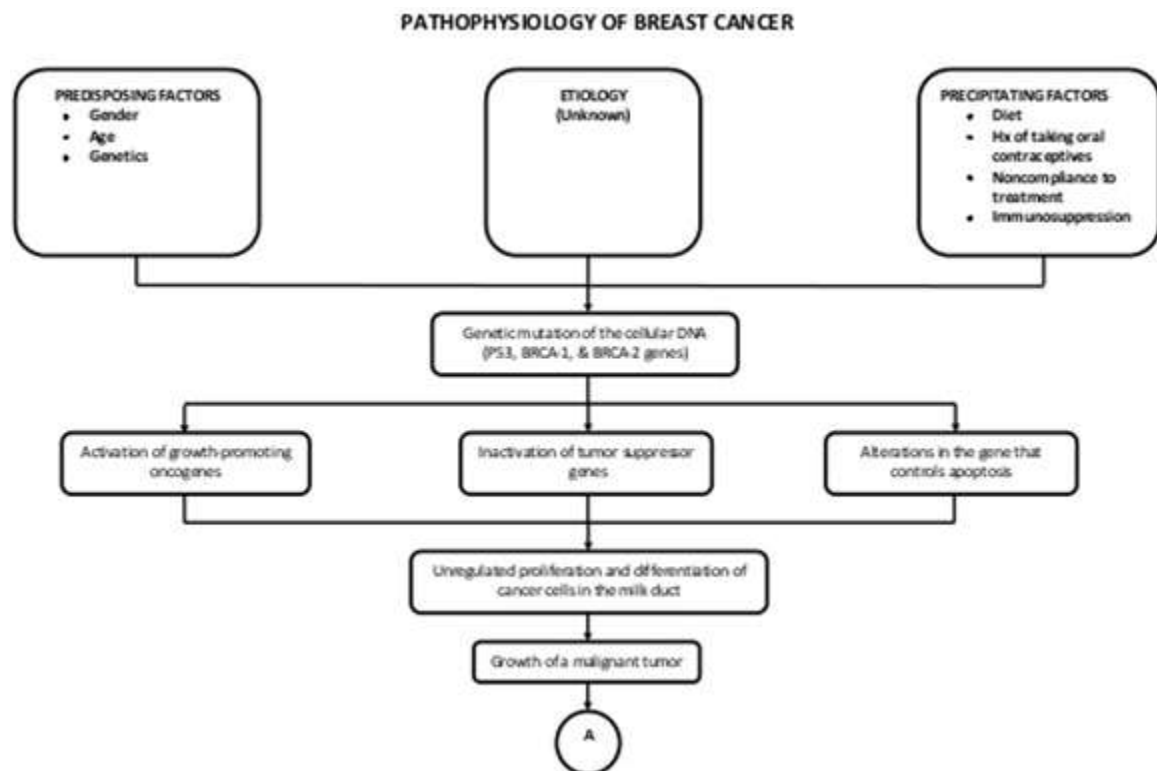


Complications of anesthesia:

During anaesthesia:	After anaesthesia:
➤ Respiratory depression	➤ Nausea and vomiting
➤ Salivation, respiratory secretions	➤ Persisting sedation
➤ Cardiac arrhythmias	➤ Pneumonia
➤ Fall in BP	➤ Organ damage – liver, kidney
➤ Aspiration	➤ Nerve palsies
➤ Laryngospasm and asphyxia	➤ Emergence delirium
➤ Awareness	➤ Cognitive defects
➤ Delirium and convulsion	
➤ Fire and explosion	

b. Describe the causes, pathophysiology and surgical management of Carcinoma Breast.
(2.5+2.5+5)

Causes and pathophysiology of Carcinoma Breast:-



Surgical management includes:-

- **Breast conservation treatment**
 - Removal of entire tumor
 - **Total mastectomy**
 - Removal of breasts, pectoral muscles, axillary lymph nodes, fat/adjacent tissue
 - **Modified radical mastectomy**
 - Remove breast and axillary nodes
 - **Sentinel node biopsy and axillary lymph node dissection**
 - **Breast reconstruction surgery**
- 28

c. A 60 year old Shilpaben Kumbhar complains of pain in both the lower limb while walking for 10 minutes which relive upon rest for 3 minutes after which she resumes her walking again. She is suffering from diabetes since past 5 years. (2.5+2.5+5)

- i. Name the clinical feature described by the patient.
- ii. Which condition patient is most likely to suffer from?
- iii. Which risk factors are associated with the development of the clinical feature described by the patient? Describe any 5 of them.

Clinical feature described by the patient is intermittent claudication. Condition which is most likely possible is Peripheral vascular (arterial) disease/ diabetic arteritis.

Risk factors are:-

1. Sex (male)
2. Age (elderly) – blood vessels become less elastic, become thin walled and calcified – □ PVR – □ BP
3. Cigarette smoking
 - nicotine causes vasoconstriction and spasm of the arteries – □ circulation to the extremities
 - CO₂ inhaled in cigarette smoke reduces O₂ transport to tissues
4. Hypertension – cause elastic tissues to be replaced by fibrous collagen tissue □ arterial wall become less distensible □ □ resistance to blood flow □ □ BP
5. Hyperlipidemia – atherosclerotic plaque
6. Obesity – places added burden on the heart & blood vessels
 - excess fat contribute to □ venous congestion

7. Lack of physical activity

- Physical activity – promotes muscle contraction
 - venous return to the heart
- aids in development of collateral circulation

8. Emotional stress – stimulates sympathetic N.S. - peripheral vasoconstriction □ □ BP

9. Diabetes mellitus – changes in glucose & fat metabolism promote the atherosclerotic process

10. Family history of atherosclerosis